MDM — Microprocessor and Microcontroller.

Mod-1

- 8086 Microprocessor
- 8086 Architecture.
- Memory segmentation.
- 8086 Pin Diagram.
- Interrupts and Interrupt processing.

Ass-1
Instructn set of 8086 Micro-Processor.

Ass-2.
I.S. of 8051 — 1st Part

Mod-2.
8086 Programming.
Addressing Modes.
Instruction set and Assember Directives.
Assembly language programming.

Mod-3.
- 8086 Interfacing.
1) clock and reset ckt of 8086.
2) 8255 Peripheral Interphase
3) functional block Diagram & Descriptions, operating modes.
System Design.
Including Memory and I/O.

Mod-4.
8051 Micro-controller. *.
Diff. MP / MC. Architecture etc, Memory organization of 8051 MC.
AM., Instruction sets,
~~Assembly lang~~ Assembly language. Programming.

Mod-5
8051 Interfacing
I/O code programe.
I/O programming.
Programming for 8051.
Serial code programming, LCD Interfacing, Stepper Motor and DC motor Interfacing

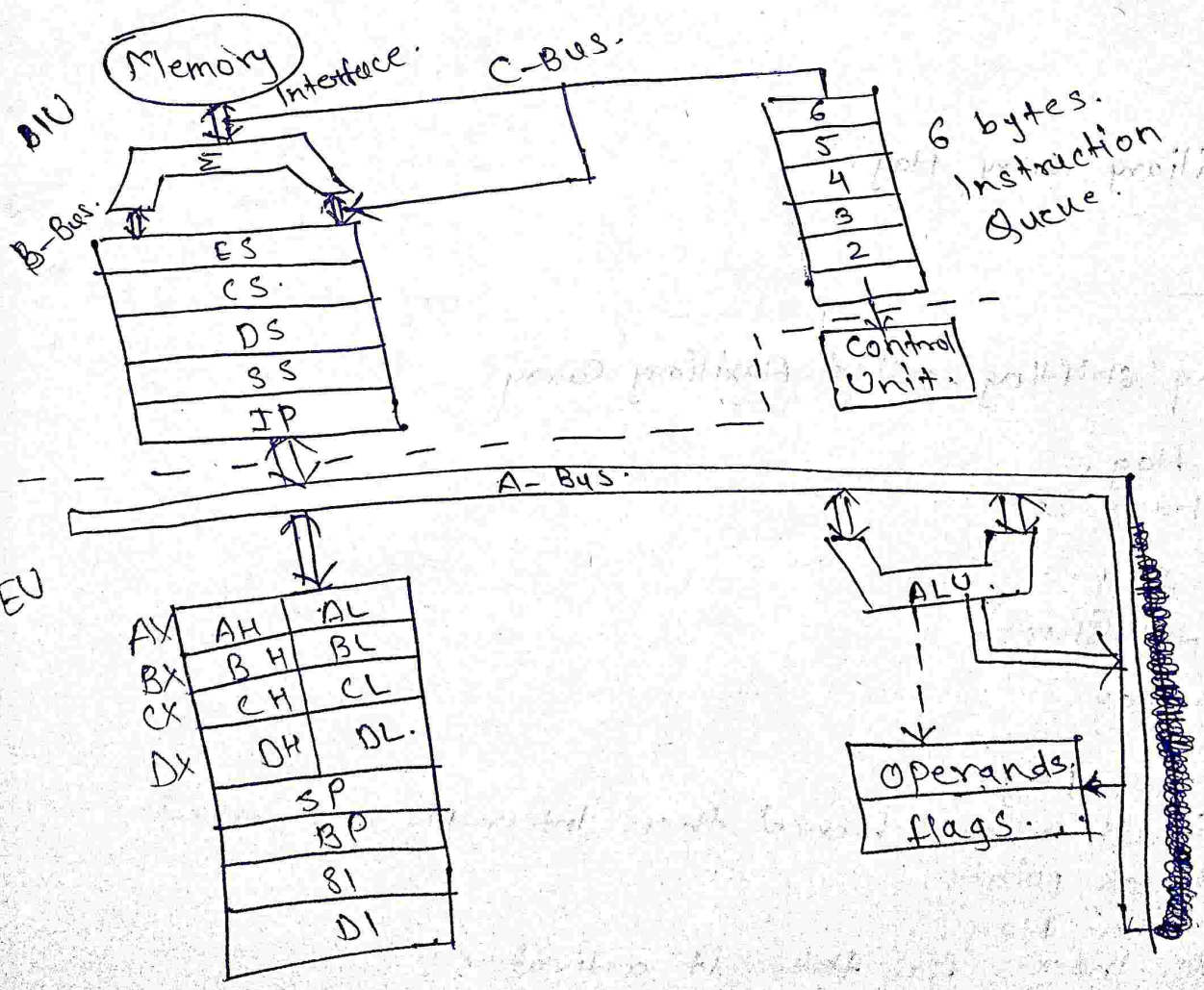
Mod-6.
Case Study on Pentium Processor. (features and Architectures only)
Case study on ARA Cortex course (" —)

Features of 8086 Microprocessor.

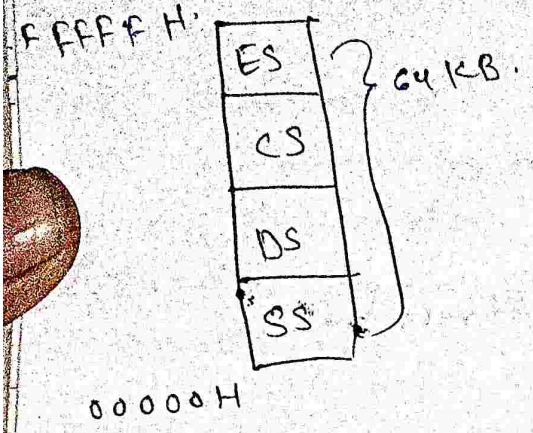
- 16 bit
- ALU size bit — 16 bit
- Size of 14 registers — 16 bit.
- It has 16 bit Data bus. and 20 bit address bus.
- We can Interface upto 1MB memory.
- 4 segment registers. — 16 bit
  Data, Code, Stack, Extra.
- Size of Each segment are $2^{16} = 64$ KB.
- It supports 3. stage pipelining.
  Fetch, Decode, Execute.
- It has 16 bit flag registers
  9 Active flag — 9
- 256 Software Interrupts.
- 5 — dedicated Interrupts.

Accumulator
Base
counter
Destination

Architecture of 8086 Microprocessor.



Segment Register :-



Flag Register :-

D15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	D0
X	X	X	X	OF	DF	IF	TF	SF	ZF	X	AF	X	PF	X	CF
														(U) undefined	

1) CF – Carry flag.
– during Binary operation.
1 – set
0 – Reset.

2) PF – parity flag.
– No. of 1's.
ex. 101101
1's – 4
Even – 1
odd – 0

3) AF – Auxiliary carry flag.
1011 1001
1001 1000

carry during shifting called Auxiliary Carry.

4) ZF – zero flag.
If answer have zero,

5) SF – Sign flag.
+ve, -ve No. (Shows)

6) TP – Trap flag.

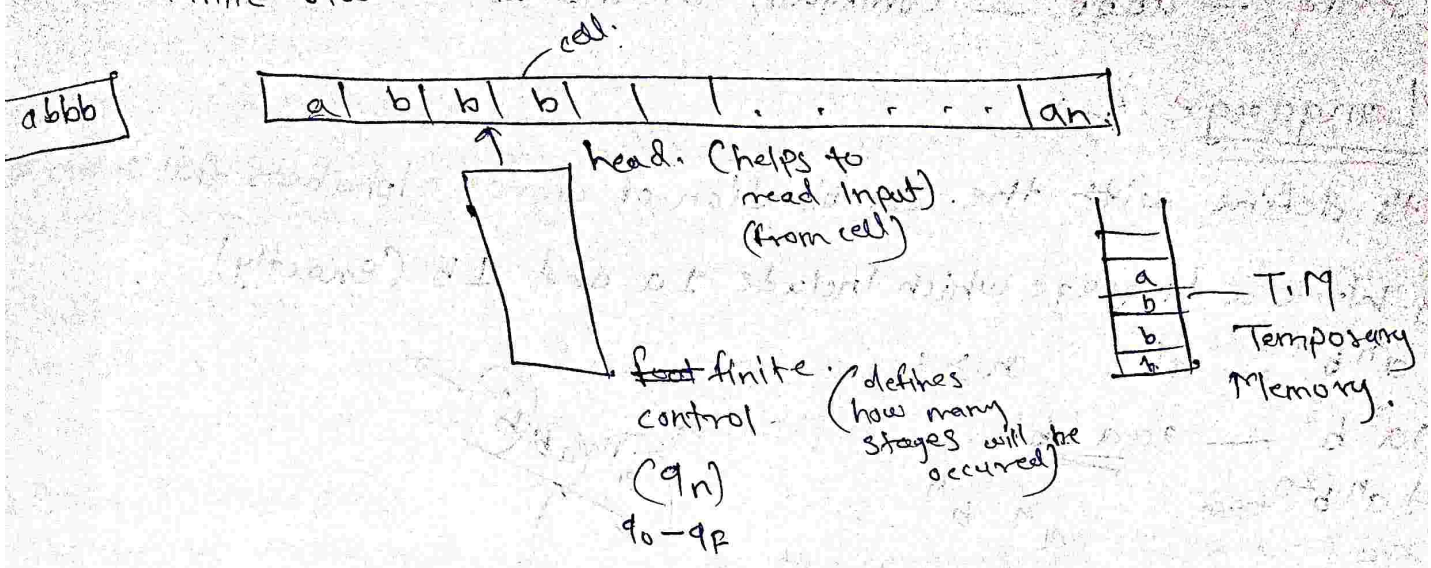
7) IF – Interrupt flag.
unwanted signals when achieved then Interrupts are called.
int sets break points.

8) DF – Direction flag.
When we use Indexes flag then it activates.

9) OF – Overflow flags.
when overflow occurs

Programmers Model of 8085 Microprocessor.

Finite State Machine.



STM

- NFA → Non Deterministic finite Automata
- DFA → Deterministic finite Automata

$\Sigma = \{a, b\}$ or $\{0, 1\}$

$\Sigma = \Sigma^0, \Sigma^1, \Sigma^2 \ldots \Sigma^n$.

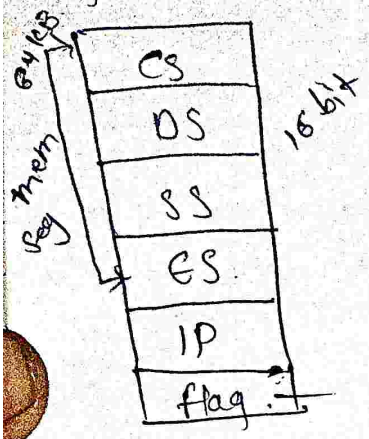
Σ^*, Σ^+

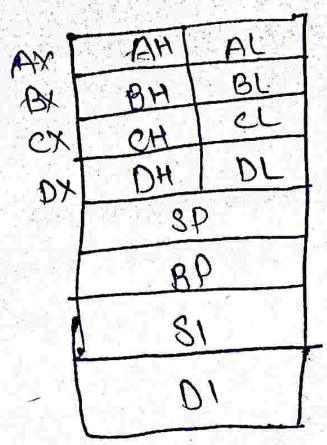
$\Sigma^* = \{\epsilon, 0, 1, 01, 11, 10, \ldots\}$. — ~~Kleen~~ Kleen closure.

$\Sigma^+ = \{0, 1, 11 \ldots\}$ — positive closure.

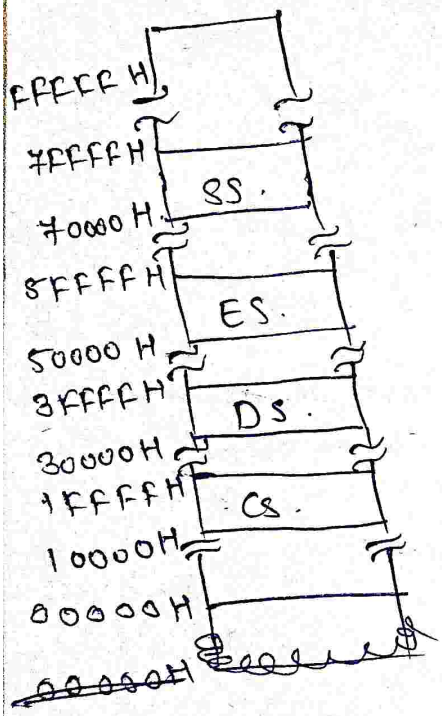
Programmers model of 8086 Microprocessor

Segment Registers.

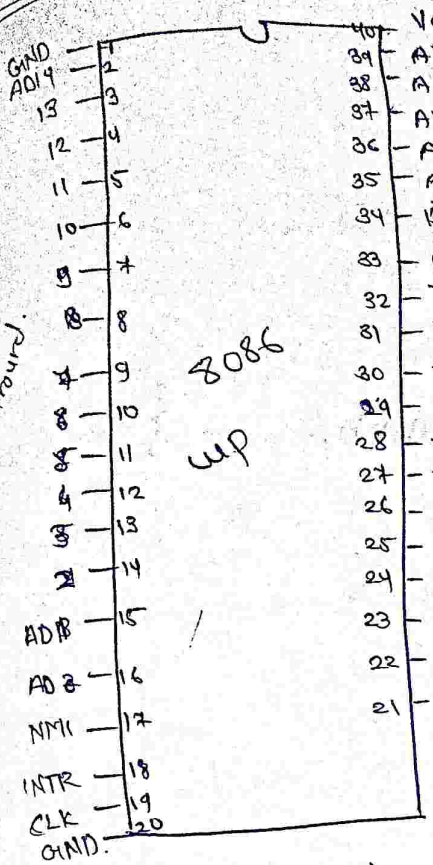




Memory Segmentation (7/8)



8086 Microprocessor Pin Diagram :-



$\overline{X}$ – active low pin.

user define – maskable
system define – non maskable.

2-16 — multiplex line.

1-20 — Gnd

~~39~~ 34-36 — multiplex line (without data).

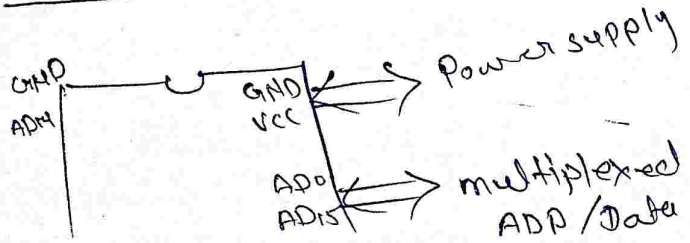
BHE — Bus High Enable.

$AD_0 - AD_{15}$ — multiplex line.

$AD_8 - AD_{15}$ — Higher order.

$AD_0 - AD_7$ — Lower order.

Functional Pin Diagram of 8086 Microprocessor :-



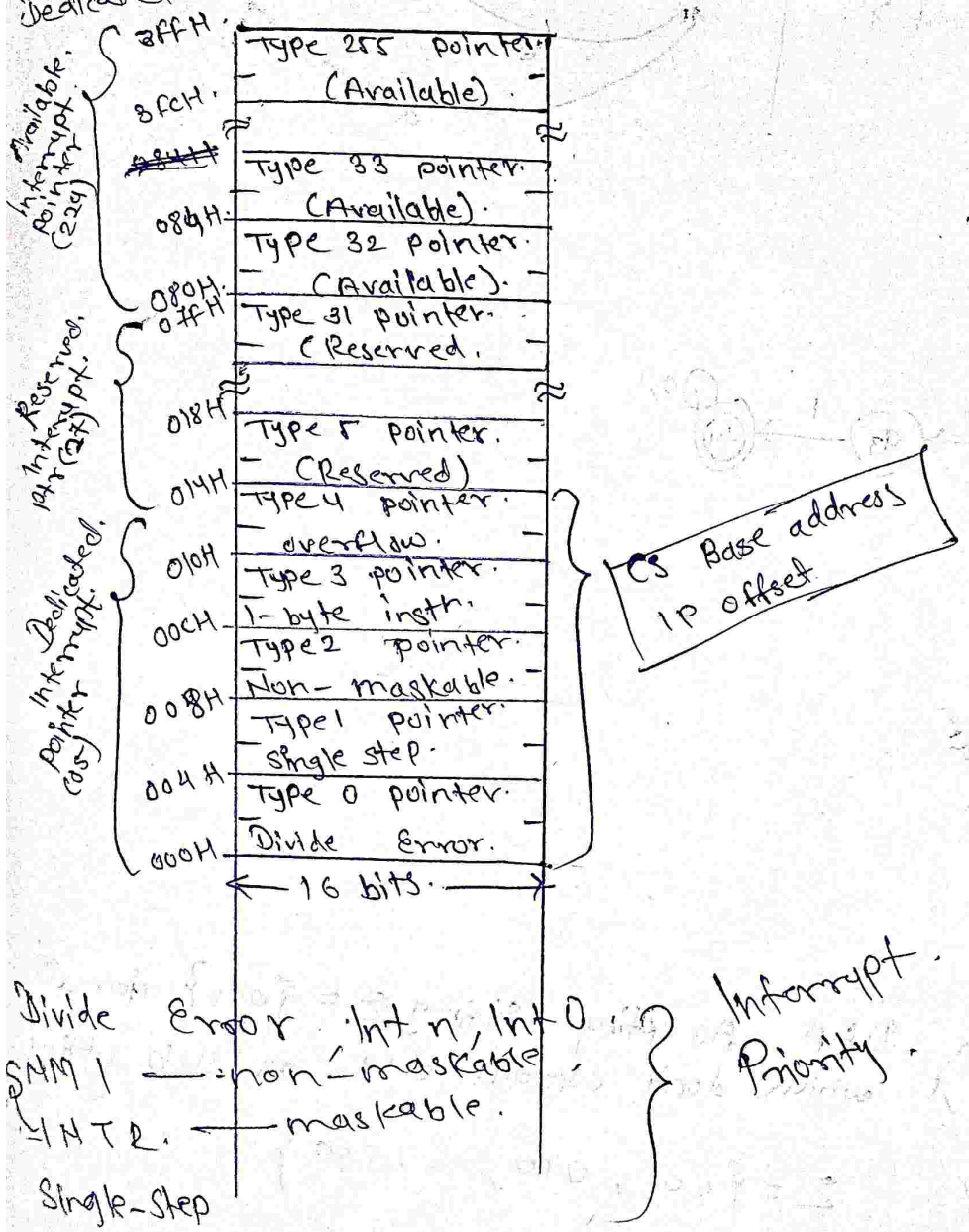
Interrupt Structure of 8086 Microprocessor

3 sources of Interrupt.

1) External.
2) Special Instructn in program.
3) Conditions produced by Instructions.

Interrupt Vector Table (IVT)

1) 256 S/w Interrupt.
Dedicated Interrupts.



Divide Error, Int n, Int O
NMI — non-maskable
INTR — maskable
Single-Step

} Interrupt Priority.

Addressing Modes of 8086 Micro-Processor.

Addressing Modes.

3 AM.

1) Data. AM
2) Programmed Memory AM
3) Stack Memory AM.

1) Data Addressing Mode.

1) register AM. (mov. ax, bx) Ex

2) Immediate AM. (move ax, 04H) Ex.

3) Memory AM — (AM for accessing Data memory)

1) Direct AM — (Registered Direct AM).

Ex mov ax, [2000 H]

2) Register Indirect AM.

LEA ax, [SI]

mov ax, [BX]

3) Based Indexed AM

mov ax, [BX+SI].

LEA , ax, [BX+SI] .

4) Register Relative AM.

mov cx, [BX+3]

5) Base relative Index AM.

mov ax, [BXSI+2]

6) String AM.

mov SB — move String Byte.

4) I/P/O/P addressing mode.

IN ax[80H].

OUT [80H], ax.

2) Programmed Memory Addressing mode

call

JMP. — non conditional.

JNZ ~~loop~~ — conditional

loop

1) Stack AM.

PUSH AX

POP AX

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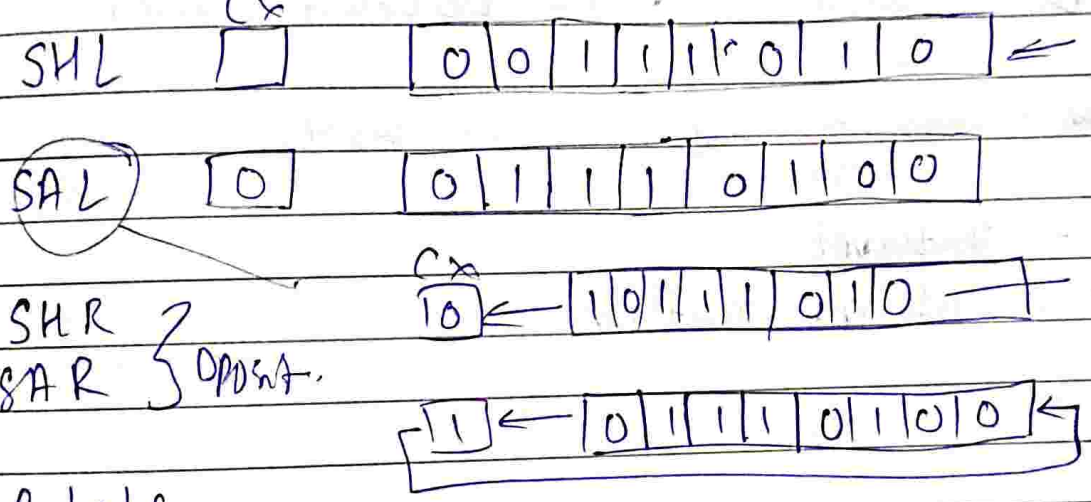
* Logical instructions.

AND :- multiplicat bx, ax
OR :- addition bx, ax
NOT :- complement cx
TEST :- AND and Com bx, ax
can't change operand value.,

* Shift and Rotate group.



* Rotate

- RCL (Rotate Left with Carry).
- ROL (Rotate Left without Carry)
- RCR (Rotate Right with carry)
- ROR (Rotate Right without Carry)

3 * String Comparing group

CMPS / CMPSB / CMPSW
Compare string
SCAS / SCASB / SCASW
Scan string string
string Byte Word

* Control Transfer Group.

JMP : we have two types of jump instructions
1. Conditional and 2. unconditional

JA / JNBE
JAE / JNB / JNC
JB / JNAE / JC

CWD - Convert ~~data to word~~ word to Double word
~~CBD~~
CBW - Convert Byte to word.

INT - Interrupt
INTO - Interrupt overflow.

MDM

Assembler ~~Directories~~ Directives :-

ALIGN

ASSUME — To assume Ex. CS: code, DS: Data, ES: Extra.

COMMENT

DB — 8

DW — 16

DD — 32

DQ — 64

DT — Data transmit

END. — Direct end

ENDS — segment end.

ENDP. — Procedure end

ENDM — Macro end.

EQU —

EVEV — Even NO.

EXITM — Exit from macro.

EXTRN — External communicath.

LABEL — loops define.

LOCAL — to use local directories

MACRO — macro activity

MODEL — short prog.

ORG. — start prog.

PAGE — define page

PROC — Procedure.

PTR. — define Label.

PUBLIC — public directories.

TITLE — Title.